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Vector Autoregressive Models

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1 Vector Autoregressive Models

1.1 Introduction and properties

Vector autoregressive (VAR) models are aimed to capture the relationship between multiple features/quantities as the change over time. For this reason VAR models are widely used in economics and life sciences.

Definition 1.1. Let $\mathbf{y}_t = (y_{1t}, \dots, y_{pt})'$ denote a vector of p distinct features observed at time, $t = 1, \dots, T$. A **vector autoregressive (VAR) process of order d** , indicated by $\text{VAR}(d)$, jointly models the levels of the p features as a stochastic function of the levels themselves, and the levels collected at the previous $t - 1, \dots, t - d$ time points. The corresponding system of stochastic equations can be written as

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \Sigma), \quad (1.1)$$

where $\mathbf{c}_t = (c_1, \dots, c_p)'$ is the the **vector of intercepts**, $\mathbf{A}_k$, $k = 1, \dots, d$, is a p -by- p (asymmetric) matrix whose elements describe the lag- k temporal interactions between the points t and $t - k$, and $\boldsymbol{\varepsilon}_t$ is a p dimensional vector of stochastic errors. Furthermore, we introduce the matrix

$$\Theta = \Sigma^{-1}, \quad (1.2)$$

called the **contemporaneous precision matrix**, which models the stochasticity between the p -features at each time point.

Note 1.1. A Vector Autoregressive model of order d has an intuitive graphical representation. At each time point t , we can define a network with set of nodes

$$V_t = \{v_1^t, \dots, v_p^t\}, \quad t = 1, \dots, T, \quad (1.3)$$

where $v_1^t, \dots, v_p^t$ are the p distinct features observed at time t , set of directed edges

$$E_d^{ts} = \{(v_i^t, v_j^s) | A_{s-t}^{ji} \neq 0\}, \quad t, s = 1, \dots, T, \quad t \neq s, \quad i, j = 1, \dots, p, \quad (1.4)$$

and set of undirected edges

$$E_u^t = \{(v_i^t, v_j^t) | \Theta_{t,ij} \neq 0\}, \quad t = 1, \dots, T, \quad i, j = 1, \dots, p, \quad i \neq j. \quad (1.5)$$

An example of such temporal network is displayed in Fig.1

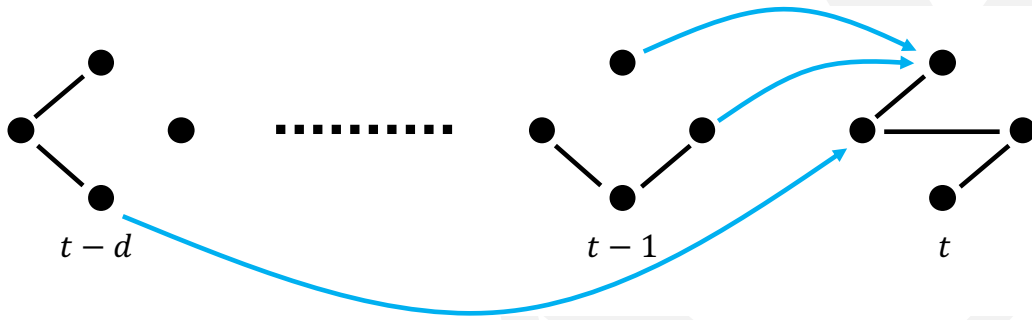


Figure 1: An example of network representation of a $\text{VAR}(d)$ model with $p = 4$ features (nodes), with contemporaneous interactions (black undirected edges) between nodes i and j weighted by $\Theta_{t,ij}$, and temporal interactions between nodes i and j within the time interval $[s, t]$ (blue directed edges) weighted by A_{s-t}^{ji} .

Definition 1.2. A stochastic process $\{X_t\}_t$ is **(weak) stationary** if its first two-order moments do not change in time.

Definition 1.3. A $\text{VAR}(d)$ model is **stable** when the roots of the **reverse characteristic polynomial**

$$\det(\mathbf{I}_p - \mathbf{A}_1 \mathbf{z} - \mathbf{A}_2 \mathbf{z}^2 - \dots - \mathbf{A}_d \mathbf{z}^d) = 0 \quad (1.6)$$

lie outside the complex unit circle, or, equivalently, if the eigen values

$$\lambda_i(\mathbf{\Gamma}), \quad i = 1, \dots, pd \quad (1.7)$$

of the **companion matrix**

$$\mathbf{\Gamma} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{d-1} & \mathbf{A}_d \\ \mathbf{I}_p & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_p & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_p & \mathbf{0} \end{bmatrix} \quad (1.8)$$

have modulus less than 1.

Proposition 1.1. *A VAR(d) model is stable if and only if there exists a constant $C > 0$ and $0 < r < 1$ such that the companion matrix $\mathbf{\Gamma}$ satisfies*

$$\|\mathbf{\Gamma}^j\| \leq Cr^j, \quad \text{for all } j \geq 0, \quad (1.9)$$

*called the **geometric decay of powers**, for some sub-multiplicative matrix norm $\|\cdot\|$, that is a norm satisfying $\|AB\| \leq \|A\|\|B\|$. The interpretation of this condition is that past shocks have less and less effect, exponentially fast.*

Proof. Let

$$\rho(\mathbf{\Gamma}) = \max_i |\lambda_i| \quad (1.10)$$

be the spectral radius of $\mathbf{\Gamma}$, i.e. the maximum absolute eigenvalue of $\mathbf{\Gamma}$, and let $\|\cdot\|$ be some sub-multiplicative matrix norm, that is satisfying $\|AB\| \leq \|A\|\|B\|$, for any pair of matrices A and B with compatible sizes. From Gelfand's formula

$$\rho(\mathbf{\Gamma}) = \lim_{k \rightarrow \infty} \|\mathbf{\Gamma}^k\|^{1/k} \quad (1.11)$$

We want to prove that

$$\rho(\mathbf{\Gamma}) < 1 \iff \exists C > 0 \quad \text{and} \quad 0 < r < 1 \quad \text{s.t.} \quad \|\mathbf{\Gamma}^j\| \leq Cr^j, \quad \text{for all } j \geq 0. \quad (1.12)$$

“ $\implies$ ”:

Assume $\rho(\mathbf{\Gamma}) < 1$ and pick any $r \in (\rho(\mathbf{\Gamma}), 1)$. From Gelfand's formula it follows

$$\lim_{k \rightarrow \infty} \|\mathbf{\Gamma}^k\|^{1/k} = \rho(\mathbf{\Gamma}) < r. \quad (1.13)$$

So, for large enough k , say $k \geq k_0$ we have

$$\begin{aligned} \|\mathbf{\Gamma}^k\|^{1/k} &< r \\ \|\mathbf{\Gamma}^k\| &< r^k. \end{aligned} \quad (1.14)$$

For small powers, $k < k_0$, define

$$C' = \max_k \frac{\|\mathbf{\Gamma}^k\|}{r^k}. \quad (1.15)$$

This means that

$$\|\mathbf{\Gamma}^k\| \leq C' r^k, \quad \forall k < k_0. \quad (1.16)$$

Therefore,

$$\|\mathbf{\Gamma}^k\| \leq Cr^k, \quad \forall k \geq 0, \quad (1.17)$$

where

$$C = \max(C', 1). \quad (1.18)$$

“ $\impliedby$ ”: Assume that

$$\exists C > 0 \quad \text{and} \quad 0 < r < 1 \quad \text{s.t.} \quad \|\mathbf{\Gamma}^k\| \leq Cr^k, \quad \text{for all } k \geq 0. \quad (1.19)$$

By taking the k -th root on both sides we obtain

$$\|\mathbf{\Gamma}^k\|^{1/k} \leq C^{1/k} r. \quad (1.20)$$

If $k \rightarrow \infty$ we obtain

$$\rho(\mathbf{\Gamma}) = \lim_{k \rightarrow \infty} \|\mathbf{\Gamma}^k\|^{1/k} \leq \lim_{k \rightarrow \infty} C^{1/k} r = r < 1, \quad (1.21)$$

and therefore the spectral radius $\rho(\mathbf{\Gamma})$ of $\mathbf{\Gamma}$ is less than 1. $\square$

Proposition 1.2. *A stable VAR(d) process*

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \Sigma), \quad (1.22)$$

is **stationary**, that is has time-invariant means and covariances, with **mean-adjusted form**

$$(\mathbf{y}_t - \boldsymbol{\mu}) = \mathbf{A}_1(\mathbf{y}_{t-1} - \boldsymbol{\mu}) + \cdots + \mathbf{A}_d(\mathbf{y}_{t-d} - \boldsymbol{\mu}) + \boldsymbol{\varepsilon}_t, \quad (1.23)$$

where

$$\boldsymbol{\mu} = (\mathbf{I}_p - \mathbf{A}_1 - \cdots - \mathbf{A}_d)^{-1} \mathbf{c} \quad (1.24)$$

is the mean of the process.

Proof. (sketch of) A VAR(d) process

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \Sigma), \quad (1.25)$$

can be rewritten as the VAR(1) process

$$\mathbf{Y}_t = \mathbf{c}^* + \mathbf{\Gamma} \mathbf{Y}_{t-1} + \mathbf{U}_t, \quad (1.26)$$

where

$$\mathbf{Y}_t = \begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-(d-1)} \end{bmatrix}, \quad \mathbf{U}_t = \begin{bmatrix} \boldsymbol{\varepsilon}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{c}^* = \begin{bmatrix} \mathbf{c} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{\Gamma} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{d-1} & \mathbf{A}_d \\ \mathbf{I}_p & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_p & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_p & \mathbf{0} \end{bmatrix}. \quad (1.27)$$

Taking the expectation of both sides of Eq. (1.26) gives

$$\begin{aligned} \boldsymbol{\mu}_t &:= E[\mathbf{Y}_t] = \mathbf{c}^* + \mathbf{\Gamma} E[\mathbf{Y}_{t-1}] + E[\mathbf{U}_t] \\ &= \mathbf{c}^* + \mathbf{\Gamma} \boldsymbol{\mu}_{t-1} + \mathbf{0} \\ &= \mathbf{c}^* + \mathbf{\Gamma} \boldsymbol{\mu}_{t-1}. \end{aligned} \quad (1.28)$$

The last equation can be recursively iterated with respect to t , starting from time $t = 0$, which gives us

$$\boldsymbol{\mu}_t = \mathbf{\Gamma}^t \boldsymbol{\mu}_0 + \sum_{j=0}^{t-1} \mathbf{\Gamma}^j \mathbf{c}^*, \quad (1.29)$$

where

$$\boldsymbol{\mu}_0 = E[\mathbf{Y}_0] \quad (1.30)$$

is the initial mean of the process. Then we notice that $\boldsymbol{\mu}_t$ converges to a fixed value as $t \rightarrow \infty$, in fact

$$\begin{aligned}
\lim_{t \rightarrow \infty} \boldsymbol{\mu}_t &= \lim_{t \rightarrow \infty} \left(\boldsymbol{\Gamma}^t \boldsymbol{\mu}_0 + \sum_{j=0}^{t-1} \boldsymbol{\Gamma}^j \mathbf{c}^* \right) \\
&= \lim_{t \rightarrow \infty} \boldsymbol{\Gamma}^t \boldsymbol{\mu}_0 + \lim_{t \rightarrow \infty} \sum_{j=0}^{t-1} \boldsymbol{\Gamma}^j \mathbf{c}^* \\
&= \mathbf{0} + \sum_{j=0}^{\infty} \boldsymbol{\Gamma}^j \mathbf{c}^* \\
&= (\mathbf{I} - \boldsymbol{\Gamma})^{-1} \mathbf{c}^*,
\end{aligned} \tag{1.31}$$

where the last equality holds because all the eigenvalues $\lambda_i(\boldsymbol{\Gamma})$ of $\boldsymbol{\Gamma}$ satisfy $|\lambda_i(\boldsymbol{\Gamma})| < 1$, and therefore $\boldsymbol{\Gamma}^t \rightarrow \mathbf{0}$ as $t \rightarrow \infty$, and the geometric series converges to $(\mathbf{I} - \boldsymbol{\Gamma})^{-1}$. Therefore, at steady the **fixed point equation**

$$\boldsymbol{\mu}_\infty = \mathbf{c}^* + \boldsymbol{\Gamma} \boldsymbol{\mu}_\infty \tag{1.32}$$

is satisfied or equivalently,

$$(\mathbf{I} - \boldsymbol{\Gamma}) \boldsymbol{\mu}_\infty = \mathbf{c}^*, \tag{1.33}$$

which gives

$$\boldsymbol{\mu}_\infty = (\mathbf{I} - \boldsymbol{\Gamma})^{-1} \mathbf{c}^*. \tag{1.34}$$

The first p elements $\boldsymbol{\mu}_\infty^y$ of $\boldsymbol{\mu}_\infty$ give the steady mean of $\mathbf{y}_t$, in fact taking the expectation of both sides of

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t \tag{1.35}$$

gives

$$E \mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 E \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d E \mathbf{y}_{t-d}, \tag{1.36}$$

which, at steady state gives

$$\boldsymbol{\mu}_\infty^y = \mathbf{c} + \mathbf{A}_1 \boldsymbol{\mu}_\infty^y + \cdots + \mathbf{A}_d \boldsymbol{\mu}_\infty^y. \tag{1.37}$$

After rearranging terms and solving for $\boldsymbol{\mu}_\infty^y$ we obtain

$$\boldsymbol{\mu}_\infty^y = (\mathbf{I} - \mathbf{A}_1 - \mathbf{A}_2 - \cdots - \mathbf{A}_d)^{-1} \mathbf{c}. \tag{1.38}$$

The mean vector $\boldsymbol{\mu}_\infty^y$ does not depend on the time t . It is worth to note that the expression for $\boldsymbol{\mu}_\infty^y$ can also be obtained as the first block (first p elements) of the full mean vector $\boldsymbol{\mu}_\infty$, by computing explicitly the right hand side of

$$\boldsymbol{\mu}_\infty = (\mathbf{I} - \boldsymbol{\Gamma})^{-1} \mathbf{c}^*. \tag{1.39}$$

Let's now define the mean-corrected process

$$\mathbf{z}_t := \mathbf{y}_t - \boldsymbol{\mu}_\infty^y, \tag{1.40}$$

and substitute $\mathbf{y}_t = \mathbf{z}_t + \boldsymbol{\mu}_\infty^y$ into the VAR(d) equation to obtain

$$\begin{aligned}
\mathbf{y}_t &= \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t \\
\mathbf{z}_t + \boldsymbol{\mu}_\infty^y &= \mathbf{c} + \mathbf{A}_1 (\mathbf{z}_{t-1} + \boldsymbol{\mu}_\infty^y) + \cdots + \mathbf{A}_d (\mathbf{z}_{t-d} + \boldsymbol{\mu}_\infty^y) + \boldsymbol{\varepsilon}_t \\
\mathbf{z}_t &= (\mathbf{A}_1 \mathbf{z}_{t-1} + \cdots + \mathbf{A}_d \mathbf{z}_{t-d}) + \underbrace{(\mathbf{c} + (\mathbf{A}_1 + \cdots + \mathbf{A}_d - \mathbf{I}) \boldsymbol{\mu}_\infty^y)}_{=0} + \boldsymbol{\varepsilon}_t \\
\mathbf{z}_t &= \mathbf{A}_1 \mathbf{z}_{t-1} + \cdots + \mathbf{A}_d \mathbf{z}_{t-d} + \boldsymbol{\varepsilon}_t
\end{aligned} \tag{1.41}$$

Therefore, $\mathbf{z}_t$ follows a zero-mean VAR(d) with the same coefficient matrices of $\mathbf{y}_t$. Recall that from the VAR(1) formulation of Eq. (1.26), we have

$$\mathbf{Y}_t = \mathbf{c}^* + \mathbf{\Gamma} \mathbf{Y}_{t-1} + \mathbf{U}_t, \quad (1.42)$$

and that the unconditional mean $\boldsymbol{\mu}_\infty = (\mathbf{I} - \mathbf{\Gamma})^{-1} \mathbf{c}^*$ satisfies

$$\mathbf{Y}_t = \mathbf{c}^* + \mathbf{\Gamma} \boldsymbol{\mu}_\infty. \quad (1.43)$$

After subtracting this from both sides of Eq. (1.26), we have

$$\begin{aligned} \mathbf{Y}_t - \boldsymbol{\mu}_\infty &= \mathbf{\Gamma}(\mathbf{Y}_{t-1} - \boldsymbol{\mu}_\infty) + \mathbf{U}_t \\ \mathbf{Z}_t &= \mathbf{\Gamma} \mathbf{Z}_{t-1} + \mathbf{U}_t. \end{aligned} \quad (1.44)$$

where we have set

$$\mathbf{Z}_t := \mathbf{Y}_t - \boldsymbol{\mu}_\infty. \quad (1.45)$$

Now, recursively substituting for $\mathbf{Z}_{t-1}$ gives

$$\begin{aligned} \mathbf{Z}_t &= \mathbf{\Gamma} \mathbf{Z}_{t-1} + \mathbf{U}_t \\ &= \mathbf{\Gamma}(\mathbf{\Gamma} \mathbf{Z}_{t-2} + \mathbf{U}_{t-1}) + \mathbf{U}_t \\ &= \mathbf{\Gamma}^2 \mathbf{Z}_{t-2} + \mathbf{\Gamma} \mathbf{U}_{t-1} + \mathbf{U}_t \\ &\vdots \\ &= \mathbf{\Gamma}^k \mathbf{Z}_{t-k} + \sum_{j=0}^{k-1} \mathbf{\Gamma}^j \mathbf{U}_{t-j}, \end{aligned} \quad (1.46)$$

which holds for any $k \geq 1$. Now, since the process $\mathbf{y}_t$ is stable, then $\mathbf{\Gamma}^k \rightarrow 0$ as $k \rightarrow \infty$, because all eigenvalues of $\mathbf{\Gamma}$ lie inside the complex unit circle (see Prop. A1.1 in the Appendix). Thus,

$$\lim_{k \rightarrow \infty} \mathbf{\Gamma}^k \mathbf{Z}_{t-k} = 0, \quad (1.47)$$

which implies

$$\mathbf{Y}_t - \boldsymbol{\mu}_\infty = \mathbf{Z}_t = \sum_{j=0}^{\infty} \mathbf{\Gamma}^j \mathbf{U}_{t-j}, \quad (1.48)$$

which is called the *infinite moving-average* (MA_∞) *representation* of the process, whose first p components are given by

$$\mathbf{z}_t = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j \boldsymbol{\varepsilon}_{t-j}, \quad (1.49)$$

where Φ_j are the first $p \times p$ block rows of A^j . Therefore, the covariance matrix of $\mathbf{z}_t$ is given by

$$\begin{aligned}
Cov(\mathbf{z}_t, \mathbf{z}_{t-k}) &= Cov \left(\sum_{j=0}^{\infty} \Phi_j \boldsymbol{\varepsilon}_{t-j}, \sum_{i=0}^{\infty} \Phi_i \boldsymbol{\varepsilon}_{t-k-i} \right) \\
&= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \Phi_j \boldsymbol{\varepsilon}_{t-j} \right) \left(\sum_{i=0}^{\infty} \Phi_i \boldsymbol{\varepsilon}_{t-k-i} \right)' \right] \\
&= \mathbb{E} \left[\sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \Phi_j \boldsymbol{\varepsilon}_{t-j} \boldsymbol{\varepsilon}_{t-k-i}' \Phi_i' \right] \\
&= \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \Phi_j \mathbb{E} [\boldsymbol{\varepsilon}_{t-j} \boldsymbol{\varepsilon}_{t-k-i}'] \Phi_i',
\end{aligned} \tag{1.50}$$

where

$$\mathbb{E} [\boldsymbol{\varepsilon}_{t-j} \boldsymbol{\varepsilon}_{t-k-i}'] = \begin{cases} \Sigma & \text{if } t-j = t-k-i \quad (j = k+i) \\ 0 & \text{otherwise.} \end{cases} \tag{1.51}$$

This means that the sum in the covariance matrix $Cov(\mathbf{z}_t, \mathbf{z}_{t-k})$ reduces to only those terms for which $j = k+i$, that is

$$Cov(\mathbf{z}_t, \mathbf{z}_{t-k}) = \sum_{i=0}^{\infty} \Phi_{k+i} \Sigma \Phi_i' \tag{1.52}$$

Now, to ensure that this expression makes sense (i.e. it has finite entries), we must check convergence of that series. Since the process is stable, we know that

$$\exists C > 0 \quad \text{and} \quad 0 < r < 1 \quad \text{such that} \quad \|\Phi_j\| \leq Cr^j, \tag{1.53}$$

that is, past shocks have less and less effect, exponentially fast. We note that

$$\begin{aligned}
\|\Phi_{k+i} \Sigma \Phi_i'\|_1 &\leq \|\Phi_{k+i}\|_F \|\Sigma\|_F \|\Phi_i\|_F \\
&\leq \|\Sigma\|_F C^2 r^i r^{k+i} \\
&= \|\Sigma\|_F C^2 r^{k+2i},
\end{aligned} \tag{1.54}$$

where $\|B\|_1 = \sum_{ij} |B_{ij}|$ and $\|B\|_F = \sum_{ij} |B_{ij}|^2$. Then, summing over all values of i , we obtain

$$\begin{aligned}
\sum_{i=0}^{\infty} \|\Phi_{k+i} \Sigma \Phi_i'\|_1 &\leq \sum_{i=0}^{\infty} \|\Sigma\|_F C^2 r^{k+2i} \\
&= \|\Sigma\|_F C^2 r^k \sum_{i=0}^{\infty} r^{2i},
\end{aligned} \tag{1.55}$$

where the series on the right-hand side is geometric, with ratio $r^2 < 1$, which therefore converges to $\frac{1}{1-r^2}$. Thus, the original series

$$Cov(\mathbf{z}_t, \mathbf{z}_{t-k}) = \sum_{i=0}^{\infty} \Phi_{k+i} \Sigma \Phi_i' \tag{1.56}$$

converges absolutely. Therefore, since $\mathbf{z}_t$ has a convergent MA_∞ representation with first two moments

$$\mathbb{E}[\mathbf{z}_t] = \mathbf{0}, \quad \text{Cov}(\mathbf{z}_t, \mathbf{z}_{t-k}) = \sum_{i=0}^{\infty} \Phi_{k+i} \Sigma \Phi_i' \quad (1.57)$$

that depend only on the lag k , but not on time t , $\mathbf{z}_t$ is stationary. Finally, since

$$\mathbf{y}_t = \boldsymbol{\mu}_\infty^y + \mathbf{z}_t, \quad (1.58)$$

where $\boldsymbol{\mu}_\infty^y$ is a constant vector, then $\mathbf{y}_t$ is also covariance stationary, with mean

$$\mathbb{E}[\mathbf{z}_t] = \boldsymbol{\mu}_\infty^y, \quad (1.59)$$

which completes the proof. $\square$

In the following we show two examples of vector autoregressive models, and how to ensure their stability, and in turn their stationarity.

Example 1.1. *Let's consider the following VAR(1)*

$$y_t = ay_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, 1) \quad (1.60)$$

In this case, the 1×1 companion matrix $\mathbf{\Gamma}$ is

$$\mathbf{\Gamma} = [a], \quad (1.61)$$

whose absolute eigenvalue is

$$|a|. \quad (1.62)$$

Therefore, stability (and in turn stationarity) of y_t is guaranteed if $|a| < 1$.

Example 1.2. *Let's consider the following VAR(1)*

$$y_t = ay_{t-1} + by_{t-2} + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, 1) \quad (1.63)$$

In this case, the 1×1 companion matrix $\mathbf{\Gamma}$ is

$$\mathbf{\Gamma} = \begin{bmatrix} a & b \\ 1 & 0 \end{bmatrix}, \quad (1.64)$$

whose eigenvalues are the roots of the characteristic polynomial

$$\begin{aligned} \det(\mathbf{\Gamma} - \lambda \mathbf{I}) &= 0 \\ \begin{vmatrix} a - \lambda & b \\ 1 & -\lambda \end{vmatrix} &= 0 \\ -(a - \lambda)\lambda - b &= 0 \\ \lambda^2 - a\lambda - b &= 0. \end{aligned} \quad (1.65)$$

Therefore, stability (and in turn stationarity) of y_t is guaranteed if

$$\begin{aligned} |\lambda_{1,2}| &< 1 \\ \left| \frac{a \pm \sqrt{a^2 + 4b}}{2} \right| &< 1. \end{aligned} \quad (1.66)$$

1.2 Parameter Inference

Consider a p -dimensional vector autoregressive process of order d ,

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_d \mathbf{y}_{t-d} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \Sigma), \quad (1.67)$$

consistent with Eq. (1.1), where $\mathbf{y}_t = (y_{1t}, \dots, y_{pt})'$ is the vector of p distinct features at time, $t = 1, \dots, T$, $\mathbf{c}_t = (c_1, \dots, c_p)'$ is the vector of intercepts, $\mathbf{A}_k$, $k = 1, \dots, d$, is a p -by- p (asymmetric) matrix whose elements describe the lag- k temporal interactions between the points t and $t - k$, and $\boldsymbol{\varepsilon}_t$ is a p dimensional vector of stochastic errors having contemporaneous precision matrix

$$\Theta = \Sigma^{-1}, \quad (1.68)$$

which models the stochasticity between the p -features at each time point. In most cases the coefficient matrices $\mathbf{A}_1, \dots, \mathbf{A}_d$, and the precision matrix Θ are unknown and have to be estimated from experimentally observed data. In the following, without loss of generality we consider that the mean vector $\mathbf{c}$ is zero, so the process is mean-corrected. In this section, a 2-step inference approach based on multivariate least squares (MLS) and maximum-likelihood (ML) is presented. More formally, the VAR model of Eq. (1.1) is first written in a multivariate regression form

$$\mathbf{Y} = \mathbf{X}\mathbf{A}' + \boldsymbol{\varepsilon}, \quad (1.69)$$

where

$$\begin{aligned} \mathbf{Y} &= [\mathbf{y}_{d+1}, \dots, \mathbf{y}_T]' \in \mathbb{R}^{(T-d) \times p}, \quad \mathbf{X} = [\mathbf{X}_{d+1}, \dots, \mathbf{X}_T]' \in \mathbb{R}^{(T-d) \times dp}, \\ \mathbf{A} &= [\mathbf{A}_1, \dots, \mathbf{A}_d] \in \mathbb{R}^{p \times dp}, \quad \boldsymbol{\varepsilon} = [\boldsymbol{\varepsilon}_{d+1}, \dots, \boldsymbol{\varepsilon}_T]' \in \mathbb{R}^{(T-d) \times p}, \\ \mathbf{X}_t &= [\mathbf{y}'_{t-1} \quad \mathbf{y}'_{t-2} \quad \cdots \quad \mathbf{y}'_{t-d}]. \end{aligned} \quad (1.70)$$

This means that, for the observation at time t we have

$$\mathbf{y}_t = \mathbf{A}\mathbf{X}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \Sigma). \quad (1.71)$$

Therefore, the log-likelihood is given by

$$\ell(\mathbf{A}, \Sigma | \mathbf{Y}) = -\frac{T-d}{2} \log |\Sigma| - \frac{1}{2} \sum_{t=d+1}^T (\mathbf{y}_t - \mathbf{A}\mathbf{X}_t)' \Sigma^{-1} (\mathbf{y}_t - \mathbf{A}\mathbf{X}_t). \quad (1.72)$$

Under Gaussian iid errors, the ML estimator of $\mathbf{A}$ is obtained by solving the optimization problem

$$\hat{\mathbf{A}}' = \underset{\mathbf{A} \in \mathbb{R}^{dp^2}}{\operatorname{argmin}} \|\mathbf{Y} - \mathbf{X}\mathbf{A}'\|_F^2 = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}, \quad (1.73)$$

that is the MLS estimator [1], where $\|\cdot\|_F$ is the Frobenius norm. Subsequently, given an estimate $\hat{\mathbf{A}}$ of the coefficient matrix $\mathbf{A}$, an unbiased sample estimate of the covariance matrix is given by

$$\hat{\Sigma} = \frac{1}{T-d} \sum_{t=d+1}^T (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)(\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' = \frac{1}{T-d} (\mathbf{Y} - \mathbf{X}\hat{\mathbf{A}})' (\mathbf{Y} - \mathbf{X}\hat{\mathbf{A}}). \quad (1.74)$$

In fact, assuming we have an estimate $\hat{\mathbf{A}}$ of the coefficient matrix $\mathbf{A}$, the conditional log-likelihood of Θ is

$$\begin{aligned}\ell(\Theta|\mathbf{Y}, \hat{\mathbf{A}}) &= \frac{T-d}{2} \log |\Theta| - \frac{1}{2} \sum_{t=d+1}^T (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' \Theta (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t) \\ &= \frac{T-d}{2} \log |\Theta| - \frac{T-d}{2} \text{tr}(S\Theta),\end{aligned}\tag{1.75}$$

where

$$S = \frac{1}{T-d} \sum_{t=d+1}^T (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)(\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)'\tag{1.76}$$

is the sample covariance matrix, and the last equality holds because

$$\begin{aligned}\sum_{t=d+1}^T (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' \Theta (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t) &= \sum_{t=d+1}^T \text{tr} \left((\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' \Theta (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t) \right) \\ &= \sum_{t=d+1}^T \text{tr} \left((\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)(\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' \Theta \right) \\ &= \text{tr} \left(\sum_{t=d+1}^T (\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)(\mathbf{y}_t - \hat{\mathbf{A}}\mathbf{X}_t)' \Theta \right) \\ &= \text{tr}((T-d)S\Theta).\end{aligned}\tag{1.77}$$

Therefore, the partial derivative of the log-likelihood with respect to Θ is given by

$$\frac{\partial}{\partial \Theta} \ell(\Theta|\mathbf{Y}, \hat{\mathbf{A}}) = \frac{T-d}{2} [\Theta^{-1} - S],\tag{1.78}$$

since from matrix calculus we know that

$$\frac{\partial}{\partial \Theta} \log |\Theta| = \Theta^{-1}, \quad \frac{\partial}{\partial \Theta} \text{tr}(S\Theta) = S.\tag{1.79}$$

Thus, setting the partial derivatives to zero gives

$$\frac{\partial}{\partial \Theta} \ell(\Theta|\mathbf{Y}, \hat{\mathbf{A}}) = \frac{T-d}{2} [\Theta^{-1} - S] = 0 \implies \hat{\Theta} = S.\tag{1.80}$$

1.3 Lag estimation and model selection

Usually, when the parameters of a VAR model are estimated it is assumed that the lag order d of the process is known. However, in practice this might not be known and therefore it must be selected properly. An objective choice can be made, for example, by employing information criteria (IC). Formally, given parameter estimates $\hat{\Theta}$ and $\hat{\mathbf{A}}$ we calculate

$$\begin{aligned}IC(d) &= C(p^*, n^*) - 2 \left[\frac{n^*}{2} \log |\hat{\Theta}| - \frac{n^*}{2} \text{tr}(S\hat{\Theta}) \right] \\ &\propto_{\Theta, \mathbf{A}} C(p^*, n^*) - n^* \log |\hat{\Theta}|,\end{aligned}\tag{1.81}$$

where

$$p^* = \frac{p}{2}(p+1) + dp^2 \quad (1.82)$$

is the degrees of freedom of the model, that is the number of non-zero parameters in $\hat{\Theta}$ and $\hat{\mathbf{A}}$, $n^* = T - d$ is the effective number of observations, and $C(p^*, n^*)$ is a regularization term that penalises the likelihood in case too many parameters are introduced, so as to avoid overfitting. So, the lag order selection is performed by computing the information criterion $IC(d)$ for a candidate set of orders, say $\{d_1, d_2, \dots\}$, and then choose d that minimizes $IC(d)$. For example, for the case of the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) [2], the formulation of $IC(d)$ reduces to

$$\begin{aligned} AIC(d) &= 2p^* - n^* \log |\hat{\Theta}| \\ BIC(d) &= p^* \log(n^*) - n^* \log |\hat{\Theta}| \end{aligned} \quad (1.83)$$

Besides selecting the best lag order d , it might be of interest to chose the most important interactions between the p features from time $(t-d)$ to t , for $t = d+1, \dots, T$. This can be translated to the problem of selecting the most important parameters, inside the coefficient matrix $\mathbf{A}$. This is also needed when there are more parameters than observations, that is in $n^* < p$ problems. One possible approach is to employ $\mathcal{L}_1$ penalised inference [2]. More formally, the optimization problem of Eq. (1.73) becomes

$$\hat{\mathbf{A}}'_\lambda = \underset{\mathbf{A} \in \mathbb{R}^{p \times dp}}{\operatorname{argmin}} \|\mathbf{Y} - \mathbf{X}\mathbf{A}'\|_F^2 + \lambda \|\mathbf{A}\|_1, \quad (1.84)$$

where $\|\cdot\|_1$ is the $\mathcal{L}_1$ norm. The smooth part $\|\mathbf{Y} - \mathbf{X}\mathbf{A}'\|_F^2$ of the objective function can be rewritten as

$$\begin{aligned} \|\mathbf{Y} - \mathbf{X}\mathbf{A}'\|_F^2 &= \operatorname{tr}((\mathbf{Y} - \mathbf{X}\mathbf{A}')'(\mathbf{Y} - \mathbf{X}\mathbf{A}')) \\ &= \operatorname{tr}(\mathbf{Y}'\mathbf{Y} - 2\mathbf{A}\mathbf{X}'\mathbf{Y} + \mathbf{A}\mathbf{X}'\mathbf{X}\mathbf{A}') \\ &= \operatorname{tr}(\mathbf{Y}'\mathbf{Y}) - 2\operatorname{tr}(\mathbf{A}\mathbf{X}'\mathbf{Y}) + \operatorname{tr}(\mathbf{A}\mathbf{X}'\mathbf{X}\mathbf{A}') \\ &\propto_{\mathbf{A}} -2\operatorname{tr}(\mathbf{A}\mathbf{X}'\mathbf{Y}) + \operatorname{tr}(\mathbf{A}\mathbf{X}'\mathbf{X}\mathbf{A}'). \end{aligned} \quad (1.85)$$

Taking the partial derivative of this expression with respect to $\mathbf{A}$ gives

$$\frac{\partial}{\partial \mathbf{A}} \|\mathbf{Y} - \mathbf{X}\mathbf{A}'\|_F^2 = -2\mathbf{Y}'\mathbf{X} + 2\mathbf{A}\mathbf{X}'\mathbf{X}. \quad (1.86)$$

The penalty term of the objective function is defined as

$$\lambda \|\mathbf{A}\|_1 = \lambda \sum_{ij} |A_{ij}| \quad (1.87)$$

has the following gradient

$$\lambda \frac{\partial \|\mathbf{A}\|_1}{\partial A_{ij}} = \begin{cases} \operatorname{sign}(\mathbf{A}_{ij}) & \text{if } \mathbf{A}_{ij} \neq 0 \\ \text{not defined} & \text{if } \mathbf{A}_{ij} = 0. \end{cases} \quad (1.88)$$

A penalised inference approach can be then derived as follows. Starting from a maximum likelihood (not penalised) estimate $\hat{\mathbf{A}}_{ML}$ of $\mathbf{A}$, a penalised estimate $\hat{\mathbf{A}}_\lambda$ can be obtained by solving the optimization problem defined by Eq. (1.84), and using $\hat{\mathbf{A}}_{ML}$ as a starting guess matrix parameter. In this work we use the box-constrained low-memory BFGS optimization method L-BFGS-B [3],

to which we provide the objective function of Eq. (1.84) and its gradient defined by Eqs. (1.86, 1.88). The box-constraints are employed as boundaries to avoid crossing the values of $\mathbf{A}$ where the gradient of the penalty term is not defined. After an estimated $\hat{\mathbf{A}}_\lambda$ for the penalised problem of Eq. (1.84) is obtained, an estimate $\hat{\Theta}_\lambda$ for the corresponding contemporaneous precision matrix is derived consistent with Eq. (1.74). The procedure is repeated for a set of combinations of different candidate values for the lag order d of the VAR process, and the penalty weight λ . The pseudocode of this procedure is reported in Algorithm 1 and it is fully implemented in R, whose code is available at [?].

Algorithm 1 Penalized Least Squares procedure

- 1: set the VAR's order d and the penalty weight λ
 - 2: choose $\mathbf{A}_0 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ as starting parameter guess
 - 3: define the penalized LS function l_p and ∇l_p according to Eqs. (1.84, 1.86, 1.88)
 - 4: compute $\hat{\mathbf{A}}_{\lambda,d} \leftarrow \underset{\mathbf{A}}{\operatorname{argmin}} -l_p(\mathbf{A}|\mathbf{Y}, \mathbf{X}, \lambda, d)$ using L-BFGS-B method
 - 5: calculate an estimate $\hat{\Theta}_{\lambda,d}$ for the precision matrix following Eq. (1.74).
-

1.4 Example

We use the inference method of Section 1.2 to analyse the the daily closing prices of four major European stock indices, with stationary VAR(d) models. The dataset **EuStockMarkets** is available on R [4]. Since we are using a stationary VAR(d) process to analyse the collected data, we need to ensure that the properties of the time series under study do not depend on the time, so that the fitted VAR(d) process is stable. However, the timeseries and the pairwise correlations between the stock prices and the time suggest that there is an impact of the time on each stock price, as displayed in the first two plot panels of Figure 2. Furthermore, a linear regression of each stock price over time suggests that homoscedasticity does not hold, as indicated by the scatterplots of residuals against fitted values displayed on the right plot panel of Figure 2. Thus, before applying the VAR(d) model, we rescale each stock's time series $\{\mathbf{y}_t\}_t$ with the following transformation

$$\mathbf{y}_t \leftarrow \log \mathbf{y}_{t+1} - \log \mathbf{y}_t. \quad (1.89)$$

Furthermore, we update each rescaled stock $\mathbf{y}_{jt}$, $j = 1, 2, 3, 4$, as

$$\mathbf{y}_{jt} \leftarrow \mathbf{y}_{jt} - \frac{1}{T} \sum_{t=1}^T \mathbf{y}_{jt} \quad (1.90)$$

so that their mean is zero, and we don't need to estimate the intercept vector $\mathbf{c}$ in Eq. (1.1). As a result, the confounding effect of the time is now removed, as suggested by the diagnostic analysis of Figure 3.

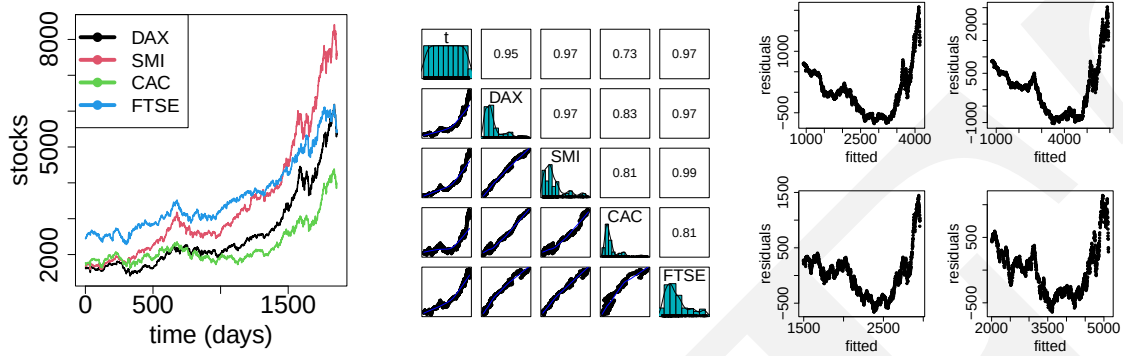


Figure 2: Left: Daily closing prices (y -axis) of major European stock indices: Germany DAX (Ibis), Switzerland SMI, France CAC, and UK FTSE over (business) time (x -axis). Center: Pairwise correlation between the stock prices and the time. Right: Residuals (y -axis) over fitted (x -axis) for a simple linear model of each stock price over time.

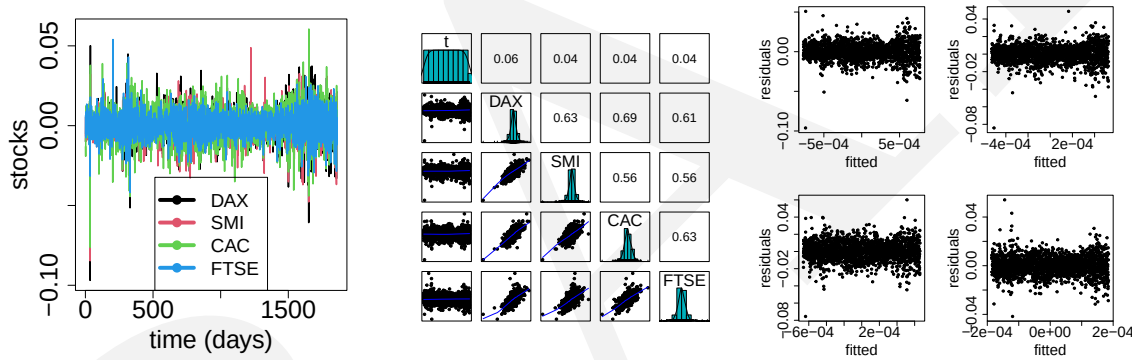


Figure 3: Left: Rescaled daily closing prices (y -axis) of major European stock indices: Germany DAX (Ibis), Switzerland SMI, France CAC, and UK FTSE over (business) time (x -axis). Center: Pairwise correlation between the rescaled stock prices and the time. Right: Residuals (y -axis) over fitted (x -axis) for a simple linear model of each rescaled stock price over time.

Using the penalised inference method of Section 1.2 and Algorithm 1, we analysed the rescaled data to estimate the matrix parameter $\mathbf{A}_{\lambda,d}$ of a $\text{VAR}(\lambda, d)$ model, under different combinations of the penalty weight λ and the VAR's order d , where $\lambda \in \{0, .001, .005, .05\}$ and $d \in \{7, 15, 30, 45\}$. Subsequently, an unbiased sample estimate $\hat{\Theta}_{\lambda,d}$ of the precision matrix is obtained consistent with Eq. (1.74). Finally, we chose the best model according to the BIC criterion. It is worth noting that $\lambda = 0$ corresponds to the multivariate least squares estimate which we use as starting point during the penalized optimization procedure.

As a result, according to the BIC selection criterion, the best combination of VAR's order d and penalty weight λ is given by $(d = 7, \lambda = 0.05)$ as shown in Figure 4.

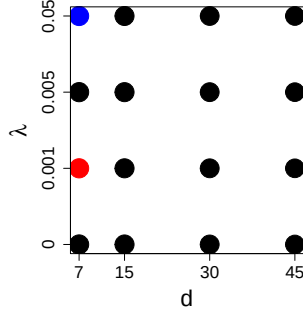


Figure 4: AIC values (dot size) for each combination of the proposed values of d and λ . The best AIC and BIC models are highlighted in red and blue respectively.

A graphical representation of the estimated transition matrices $\mathbf{A}_1, \dots, \mathbf{A}_7$ is given in Figure 5, together with the estimated contemporaneous covariance matrix $\hat{\Theta}_{\lambda,d}$. Looking at the transition matrices $\mathbf{A}_5, \mathbf{A}_4, \mathbf{A}_1$ it seems that the stocks S, D and F are mainly guiding the behaviour of the market prices with a time lag of 5, 4 and 1 respectively. It seems also that the price of stock F at time point t is affected by it self from the previous time point $t - 1$. The remaining transition matrices do not show any evidence of a significant impact between the stock prices for the time lags considered in the analysis, thus suggesting that VAR(d) processes with lag order d lower than 7 could be considered as potential alternative models. Furthermore, the estimated contemporaneous precision matrix $\hat{\Theta}_{\lambda,d}$ suggests a (conditional) correlation between stocks C, S, D given F at each time-point.

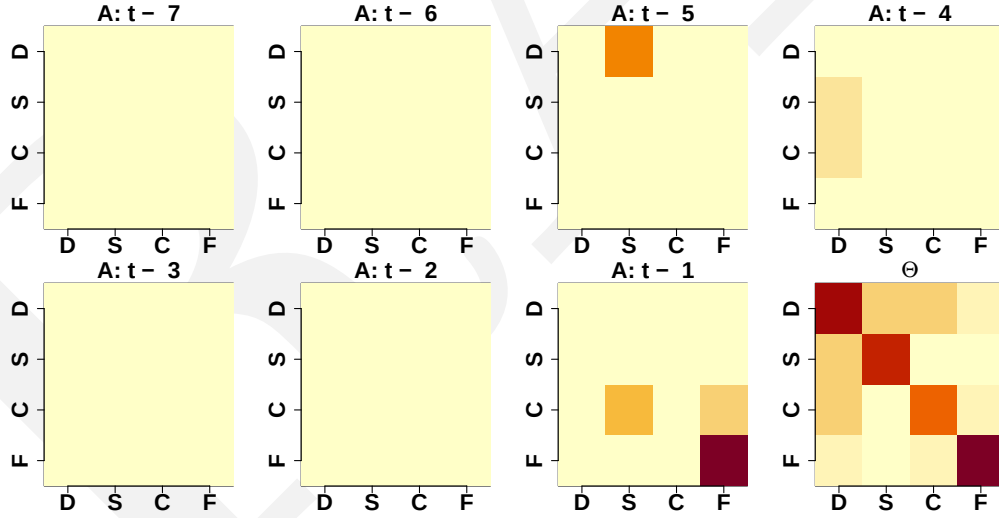


Figure 5: Transition matrices $\mathbf{A}_7, \dots, \mathbf{A}_1$ corresponding to the estimate $\hat{\alpha}_{d=7, \lambda=0.05}$. There is an impact from i -th stock to j -th stock from time point $t - d$ to time point d if $\mathbf{A}_{d,ji} \neq 0$. The darker the color the greater the impact (in absolute value).

□

Appendices

A1 Results from linear algebra

Proposition A1.1. *If all eigenvalues λ_i of a matrix A satisfy $|\lambda_i| < 1$, then $A^j \rightarrow 0$ as $j \rightarrow \infty$.*

Proof. The Jordan decomposition (or the Schur decomposition in case of non-diagonalizable matrices) gives

$$A = PJP^{-1} \tag{A1.1}$$

which raised to the power of j gives

$$A^j = PJ^jP^{-1}, \tag{A1.2}$$

where J is the Jordan form (diagonal or block-diagonal), and its entries (or blocks) are raised to the power of j . Now, since all $|\lambda_i| < 1$ we have that $\lambda_i^j \rightarrow 0$ as $j \rightarrow \infty$, each Jordan block tends to zero, $J^j \rightarrow 0$ as $j \rightarrow \infty$, and therefore $A^j = PJ^jP^{-1} \rightarrow 0$, as $j \rightarrow \infty$. $\square$

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